

Minerva SetUp and WalkThroughDocument

Hi there! This tutorial will run you through a basic classification task in Minerva using Jupyter Notebook

There are some pre-requisites before attempting this:

1. Read the [README.md](https://git.chalmers.se/karppa/minerva) on the Minerva cluster on this page:
<https://git.chalmers.se/karppa/minerva>
 - i. Thank you Matti for such lovely documentation! ✨
 - ii. Get access to the cluster by following the instructions outlined in the document, under "Getting access"
2. Read the [QUICKSTART.md](https://git.chalmers.se/karppa/minerva/-/blob/main/QUICKSTART.md?ref_type=heads) on the Minerva cluster on this page:
https://git.chalmers.se/karppa/minerva/-/blob/main/QUICKSTART.md?ref_type=heads
 - i. Thank you Matti for such lovely documentation! ✨
3. Now that you've read the documents linked in points 1 and 2, let's get cracking!

PSA: There are 2 ways to Run Jobs on Minerva

Option A: Jupyter Notebook Style  **OR Option B: Pure Python Style** 

   The **Option B** is the **ONLY** way to have your script run on GPU. If you want to use the GPU and make the most use of the cluster, please do Option B!

   If you want to stick to Jupyter Notebooks (**Option A**), all fine but it **will not** run on GPU as there is no GPU access for JupyterHub. Please decide accordingly.

Now that we understand **Option A = NO GPU** (😞) and Option B = GPU (😊), let's overview how to get set-up for each Option!

Option A: Jupyter Notebook Style Set Up

1. Open a Jupyter Notebook as per instructions:

JupyterHub

If you want to do interactive processing with Jupyter Notebooks, you can log in directly to one of:

- <http://minerva.cse.chalmers.se/jupyter>
- <http://minerva.cse.chalmers.se/saturn>

Use your CID and password to log in. As before, access is only available from Chalmers network, and to access the server from outside networks requires the use of VPN. Note that there is no GPU access for JupyterHub.

 Make a jobscript as per instructions, in your minerva terminal:

(in terminal) nano. We do this ONLY so we can access any special libraries and packages you may (have) install(ed) in your conda/venv environment.

```
nano <NAME_YOUR_JOBSCRIPT>.sh
```

Type into your jobscript:

```
#!/bin/bash
#SBATCH -c 1
#SBATCH --mem=1G
#SBATCH -t 0:30:00

source /data/users/<CID>/<NAME_OF_YOUR_ENV>/bin/activate
python3 /data/users/<CID>/<NAME_OF_YOUR_PYTHON_JOB>.py
jupyter lab --config="${CONFIG_FILE}"
```

 CTRL + X to save it

 Y to confirm jobscript name

Done!

Now in our minerva terminal, activate your venv/conda environment

 You should have made a venv/conda when you read the [QUICKSTART.md](https://git.chalmers.se/karppa/minerva/-/blob/main/QUICKSTART.md?ref_type=heads) on the Minerva cluster on this page: https://git.chalmers.se/karppa/minerva/-/blob/main/QUICKSTART.md?ref_type=heads

 If you didn't, go back and do that now!  

Back with us? Welcome back.

Now, with your venv/conda activated, type in your minerva terminal:

```
python -m ipykernel install --user --name=YOUR_ENV_NAME --display-name="SOMETHING"
```

This enables us to use all of the packages we install in our venv/conda environment in the Jupyter Notebook server 🌟

⚠️🔔🔔 REMEMBER! We're not on the GPU, we're just using our conda/venv to access our libraries.

🌟 Restart your kernels in the Jupyter server:

The screenshot shows the Jupyter Notebook interface. The 'Kernel' menu is open, displaying options such as 'Interrupt Kernel', 'Restart Kernel...', 'Restart Kernel and Clear Outputs of All Cells...', 'Restart Kernel and Run up to Selected Cell...', 'Restart Kernel and Run All Cells...', 'Restart Kernel and Debug...', 'Reconnect to Kernel', 'Shut Down Kernel', 'Shut Down All Kernels...', and 'Change Kernel...'. The 'Change Kernel...' option is highlighted. The background shows a code cell with the following Python code:

```
[1]: # Sentiment Classification

# SETUP
import pandas as pd
import numpy as np
import os

# Create figures directory if it doesn't exist
if not os.path.exists('figures'):
    os.makedirs('figures')

# LOAD DATA PATHS
# USE TAB SEPARATOR BECAUSE OF COMMA USAGE IN THE TEXT
crowd_train = pd.read_csv(r'/data/users/kelsey/DAT341/PA3/textclass_data/crowdsourced_train.csv', sep='\t')

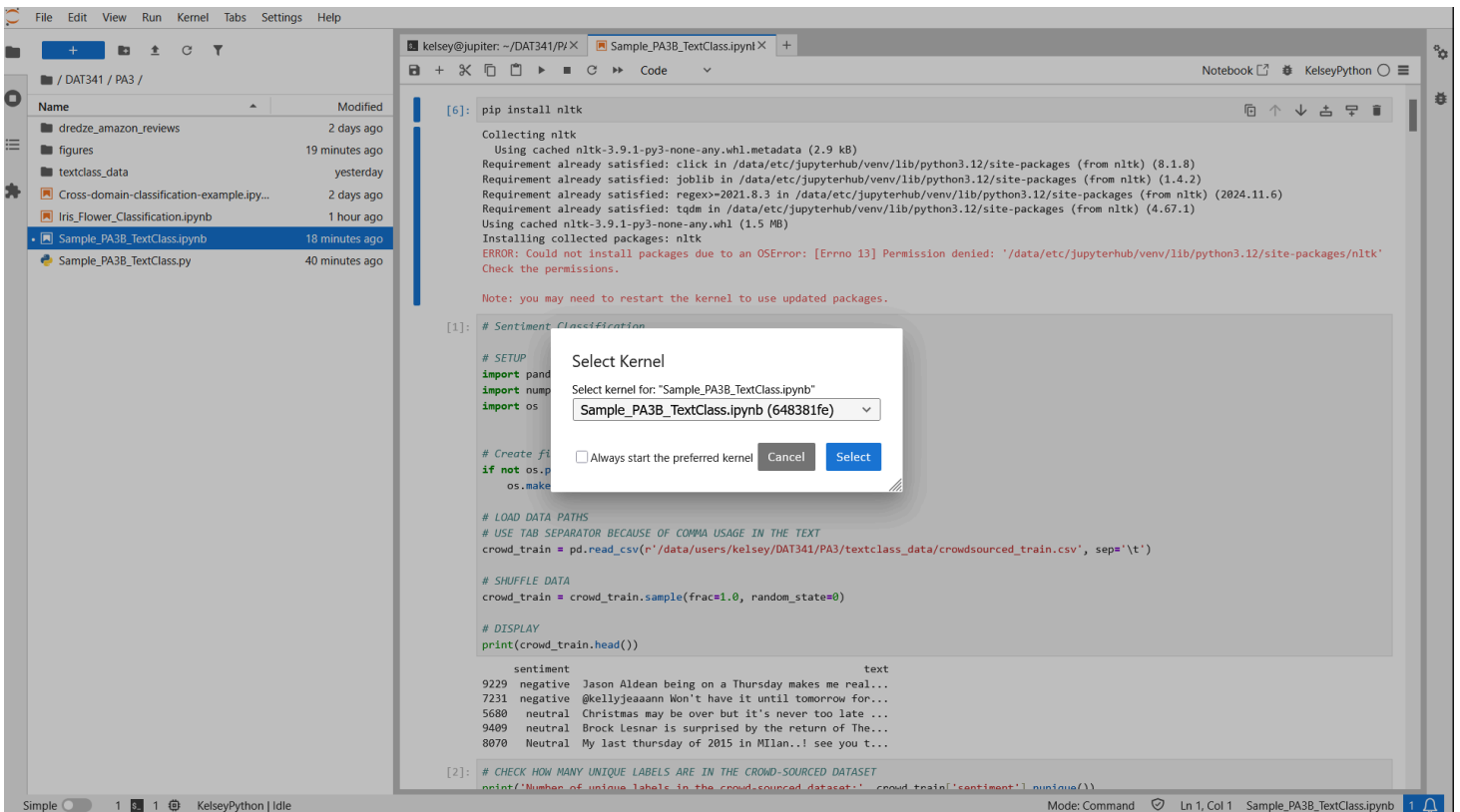
# SHUFFLE DATA
crowd_train = crowd_train.sample(frac=1.0, random_state=0)

# DISPLAY
print(crowd_train.head())

      sentiment      text
9229  negative  Jason Aldean being on a Thursday makes me real...
7231  negative  @kellyjeaann Won't have it until tomorrow for...
5680  neutral  Christmas may be over but it's never too late ...
9409  neutral  Brock Lesnar is surprised by the return of The...
8070  Neutral  My last thursday of 2015 in Milan..! see you t...

[2]: # CHECK HOW MANY UNIQUE LABELS ARE IN THE CROWD-SOURCED DATASET
print('Number of unique labels in the crowd-sourced dataset:', crowd_train['sentiment'].nunique())
```

Now you should be able to select the kernel you just made (in drop-down menu)



Congratulations! You should now be able to run your Jupyter Notebooks in your own kernel, using the packages you install(ed) in your venv/conda environment. Again, this does not have GPU access!

Option B: Pure Python Style Set Up

The set-up for Option B will borrow components from Option A. Mainly setting up a job-script, where we will specify:

- Our conda/venv environment
- Information about run-time
- Information about GPUs we need for the job
- Our jobscrip location

 I presume that you already have a basic python script that you will use for this task.

 Make a jobscrip as per instructions, in your minerva terminal:

(in terminal) nano. We do this ONLY so we can access any special libraries and packages you may (have) install(ed) in your conda/venv environment.

```
nano <NAME_YOUR_JOBSCRIPT>.sh
```

Type into your jobscrip:

```
#!/bin/bash
#SBATCH -c 1
#SBATCH --mem=1G
#SBATCH -t 0:30:00

source /data/users/<CID>/<NAME_OF_YOUR_ENV>/bin/activate
python3 /data/users/<CID>/<NAME_OF_YOUR_PYTHON_JOB>.py
jupyter lab --config="${CONFIG_FILE}"
```

- 👉 CTRL + X to save it
- 👉 Y to confirm jobscript name

Done! You're now ready to run your job in the Minerva terminal:

```
sbatch <NAME_YOUR_JOBSCRIPT>.sh
```